

enabling high-speed connectivity. The module provides cost-effective wireless gigabit connectivity for edge access solutions at a significantly lower TCO. It supports a data rate



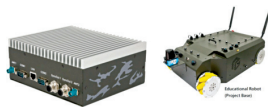
of up to 3.6Gbps with a 1.8Gbps uplink and 1.8Gbps downlink. The client node supports up to 7.6Gbps with channel bonding for point-to-multipoint and point-to-point configurations. The device supports TDMA/TDD channel access and BPSK/16-QAM modulation and coding schemes. It features an antenna with a gain of 40.5/44.5dBi with auto-beamforming. This radio has an EIRP of up to 60.5dBm and a latency of less than 1ms. The system works using a cloud based or on-premise software platform, cnMaestro management system, for secure, end-to-end network control.

Cambium Networks

<https://www.cambiumnetworks.com>

Autonomous robot controller

Aaeon RBX-I2000 is an autonomous robot controller with an integrated hardware time synchronisation. It supports an external nine-axis inertial measurement unit, which gives it an exceptional speed and positional accuracy.



It has four ports for Ethernet, two channels of CAN, 8-bit digital IO, a Fakra Z antenna port for the optional M2 Wi-Fi and Bluetooth card, and Fakra C for the GNSS antenna. It has power necessary for hardware based time synchronisation and can facilitate multiple sensor synchronisations. The RBX-2000 can be used for applications such as smart agriculture and smart mining.

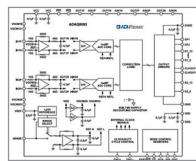
AAEON

<https://www.aaeon.com>

Multichannel DAQ system

ADAQ8092 from ADI is a 14-bit MSPS µModule with a high-speed dual-channel data acquisition (DAQ) solu-

tion. The multichannel DAQ system is a signal chain sub-system interfaced to multiple inputs (typically sensors) with an intent of converting the analogue signal into digital



data that can be comprehended by a processing unit. It can simplify the development of high-speed data acquisition systems by transferring the design burden, component selection, optimisation, and layout from the designer to the device. The device features signal conditioning, an analogue-to-digital driver, a voltage reference, and an ADC in a single package. Built-in power supply decoupling capacitors enhance power supply rejection performance, making it a robust DAQ solution. Applications include communication systems, cellular base station systems, human-computer interfaces, etc.

Analog Devices Inc.

<https://www.analog.com/en/index.html>

SOFTWARE AND TOOLS

Unified IDE for RISC-V platform

Intel Pathfinder is a unified integrated development environment (IDE) launched for supporting the wider adoption of the RISC-V platform. It is an open source platform aimed at both individual and commercial developers,



which can save time in assembling and testing different IP combinations in a single environment. The Pathfinder can simplify developing products using RISC-V cores and IP from the partner ecosystem, and instantiate on FPGA and simulator platforms. It enables pre-silicon developers to write, run, debug, and port software code on top of Intel-provided operating system stacks. Moreover, it has the ability to run industry-leading operating systems and toolchains within a unified IDE. The IDE comes

with an industry-standard toolchain and robust software.

Intel

<https://www.intel.com>

Battery pack design platform

IONETIC's ARC is a customisable software based platform that can assist battery design engineers in optimising their battery pack. This



platform can reduce the design time and cost by simplifying system design, which includes

homologation, embedded control, manufacturing options, and vehicle integration, to name a few. The tool is said to enhance the energy density in an existing design by up to 30%. The bespoke design platform is capable of generating a customised battery pack concept in minutes—automatically analysing all relevant parameters to optimise the solution for each customer's needs.

Ionetic

<https://ionetic.uk>

MISCELLANEOUS

Cost-effective stencil cleaners

KYZEN has released three cost-effective stencil cleaners, CYBERSOLV C8882, E5631, and E5631J. The E5631 and E5631J are capable of cleaning all types of solder



paste flux from stencils and A-side misprints without damaging them. Moreover, the cleaners rinse easily without leaving a trace, thus reducing the dry time. While the CYBERSOLV is a solvent based stencil cleaning fluid, all the cleaners can be used for hand-wipe and even in advanced cleaning processes, such as automated under-stencil wipe processes, ultrasonic cleaning systems as well as spray systems designed for solvent.

Kyzen

www.kyzen.com